B.Tech. CSE (III YEAR - VI SEM)

(2025-2026)

DEPARTMENT OF COMPUTER ENGINEERING & APPLICATIONS

GLA UNIVERSITY MATHURA

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Mall-Mind: The Hyper-Personalised Agentic Shopping Concierge

Mentor Name: Dr Ruby Panwar

Signature: Panwar 5/2/26

1. Overview

Problem Statement:

With the rapid growth of e-commerce platforms, physical shopping malls are losing customer engagement due to the lack of personalization and digital intelligence. Mall visitors often feel overwhelmed by the large number of stores and are unable to quickly identify relevant shops, available sizes, ongoing discounts, or optimal navigation paths. Unlike online platforms, physical malls do not collect or utilize session-level visitor data, leading to poor shopping experiences and reduced conversions.

Goal:

To design and develop a **mobile-first intelligent shopping assistant** that uses **Agentic AI** to personalize the in-mall shopping experience by providing real-time store recommendations, inventory checks, smart navigation, and dynamic offers.

Non-Goals (v1):

- Online payment gateway integration
- Nationwide mall network deployment
- Native mobile application (Android/iOS)

Value Proposition:

- **Shoppers**: Tailored recommendations, real-time inventory checks, and optimised walking routes.
- **Retailers**: Higher conversion rates through dynamic discounting and better understanding of foot traffic data.
- **Mall Management**: Increased "dwell time" and digitised insights into visitor behaviour.

2. Scope and Control

2.1. In-scope:

- QR-based mall entry and session initiation
- Natural language shopping request input
- Personalised store and product recommendations

- Real-time inventory lookup (simulated store APIs)
- Indoor navigation with optimised walking paths
- Dynamic discount generation
- Virtual try-on using AI image generation
- Interactive mall map visualisation

2.2. Out-of-scope:

- Real-world payment transactions
- Third-party loyalty program integration
- Hardware-based AR devices

2.3. Assumptions:

- Mall stores expose basic inventory data (simulated)
- Users have smartphones with internet access

2.4. Constraints:

- Academic timeline (12 weeks)
- Limited real-world store integrations
- Use of free or student-tier APIs

2.5. Dependencies:

- OpenAI Assistants API
- Mapbox / Indoor mapping library
- Node.js runtime environment
- Cloud database (MongoDB / PostgreSQL)

2.6. Acceptance Criteria:

- Personalised route generated within 5 seconds
- Inventory matching accuracy ≥ 85%
- Mentor approval of system design and final demo

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirement Analysis	Isha	Isha	Mentor	Team
System Design	Ishita	Isha	Mentor	Team
Backend Development	Prerita	Isha	Mentor	Team
Frontend Development	Ishita	Isha	Mentor	Team
AI Agent Development	Prerita	Isha	Mentor	Team
Testing & Validation	All Members	Isha	Mentor	Team
Documentation & Final Submission	Isha	Isha	Mentor	Department

4. Team and Roles

Member Name	Role	Responsibilities	Key Skills
Isha	Team Lead & Backend Developer	Requirement gathering, system coordination, backend API design, final integration	Node.js, APIs, System Design
Ishita	Frontend Developer	UI/UX design, PWA development, interactive mall maps, frontend-backend integration	React, UI/UX Design
Prerita	AI & Data Engineer	AI agent logic, NLP processing, recommendation flow, database design	AI Concepts, LLMs, Databases

5. Week-wise Plan

Week	Milestone	Isha (Lead / Backend)	Ishita (Frontend / Telehealth)	Prerita (AI Engineer)	Deliverable
1–2	Planning & Research	Requirement gathering, API scope	UI/UX flow brainstorming	AI use-case study (scheduler, reminders, renewals)	Finalized requirements & schema
3–4	System Design	API blueprint, agent workflow map	Wireframes, dashboard layout	AI prototype plan (slot + reminder logic)	Design approval & prototype plan
5–6	Backend Development	Auth, agent APIs, event flow	–	API hooks for AI integration	Backend + API demo
7–8	Frontend Development	API integration (appointments, reminders)	Dashboards, Telehealth UI (WebRTC)	–	Full-stack demo (basic)
9–10	AI & Automation	Backend support for AI endpoints	Reminder UI, adaptive rescheduling integration	Slot optimization, No-show prediction,	AI demo (smart scheduling + reminders)

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11–12	Testing & Documentation	Backend bug fixes, load/security testing	UI polish, usability testing	Agent workflow testing, explainability validation	Final submission

6. Users and UX

Personas:

- **Shopper**: Wants fast, relevant, stress-free shopping
- **Store Manager**: Wants higher engagement and sales
- **Mall Admin**: Wants analytics and smoother operations

Top user journey:
User → Scan QR → Enter shopping intent → Agent recommendations → Optimised navigation → Discounts → Purchase.

7. Market & Competitors

Platform / System	Features Offered	Limitations	Mall-Mind Differentiator
Amazon / Flipkart	Personalised recommendations, dynamic pricing	No physical experience, no in-store navigation	Combines personalisation with real-world shopping
Traditional Mall Apps	Store listings, static maps, offers	No AI, no real-time inventory, no personalisation	AI-powered recommendations and live routing
Google Maps (Indoor)	Basic indoor navigation	No shopping intent awareness	Intent-based optimised shopping paths
Brand-	Product discovery within	Limited to a single store	Cross-store, mall-wide

Specific Apps	one brand		intelligence

8. Objectives & Success Metrics

- O1: Reduce shopping time by ≥30%
- O2: Increase store visit relevance by ≥40%
- O3: Dynamic offers improve conversion by ≥20%
- O4: Navigation accuracy ≥95%

9. Key Features

- **Intelligent User Onboarding:**
  Users initiate their shopping session by scanning a QR code at the mall entrance. This creates a temporary digital session without requiring app installation. Users can then enter their shopping intent using natural language, such as product type, color, budget, and preferences.
- **Inventory Scout Agent:**
  The Inventory Scout Agent communicates with simulated store inventory APIs to identify products that match the user's requirements. It checks real-time availability, sizes, colors, and price ranges across multiple stores within the mall, ensuring that only relevant stores are recommended to the user.
- **Smart Indoor Navigation:**
  Mall-Mind provides an interactive indoor navigation system using a digital mall map. The Navigator Agent calculates the most efficient walking route between selected stores and amenities, minimizing distance and time. The route dynamically updates based on changes such as store crowd levels or closures.
- **Personal Stylist Agent:**
  This agent analyses the user's input using Natural Language Processing (NLP) to extract preferences, including style, fit, occasion, and budget. Based on historical data (if available), it refines recommendations to match the user's taste, similar to an online shopping recommender system.
- **Virtual Try-On Integration**
  The platform includes a virtual try-on feature that lets users upload a selfie to see

how a selected outfit might look on them. This feature uses Generative AI to simulate clothing fit and appearance, increasing user confidence before making a purchase decision.

10. Architecture (High-Level)

- **Frontend:** PWA (Next.js / React)
- **Backend:** Node.js + GraphQL
- **Database:** MongoDB / PostgreSQL
- **AI Layer:** OpenAI Assistants API
- **Mapping:** Mapbox Indoor SDK
- **Telehealth:** WebRTC / Jitsi

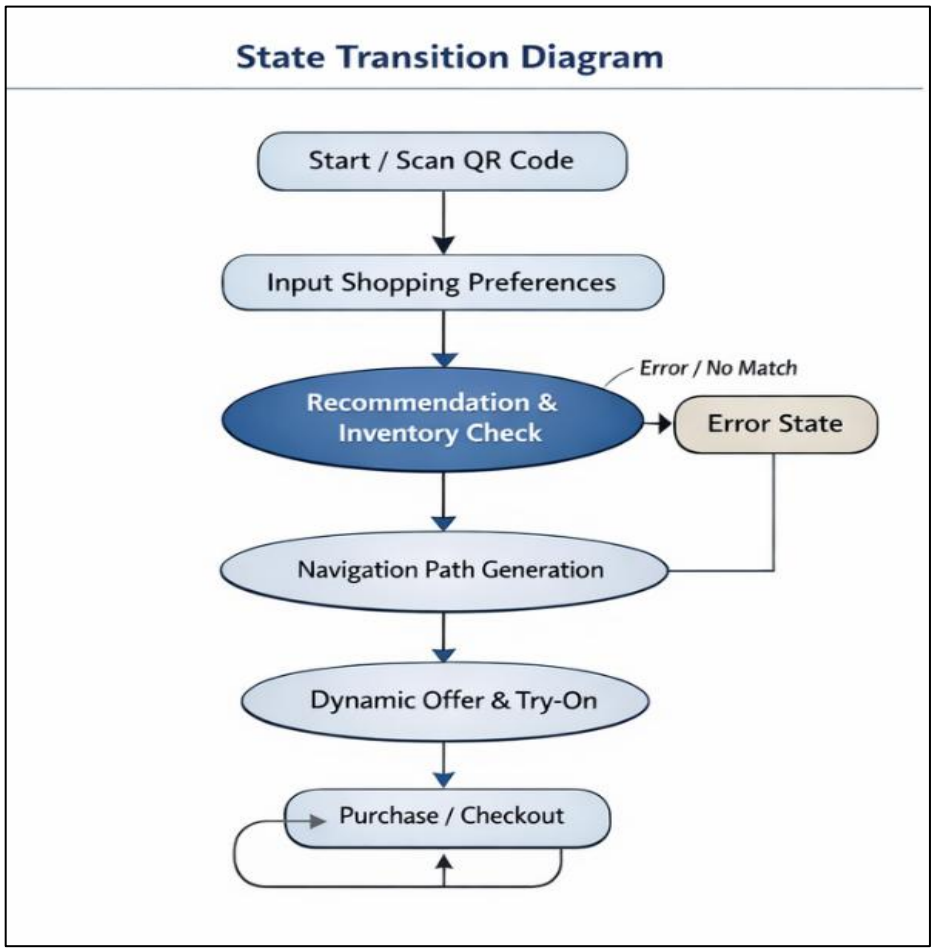
11. Data Design

To support the "Mall-Mind" ecosystem, the database must track the relationship between users, stores, and the agents.

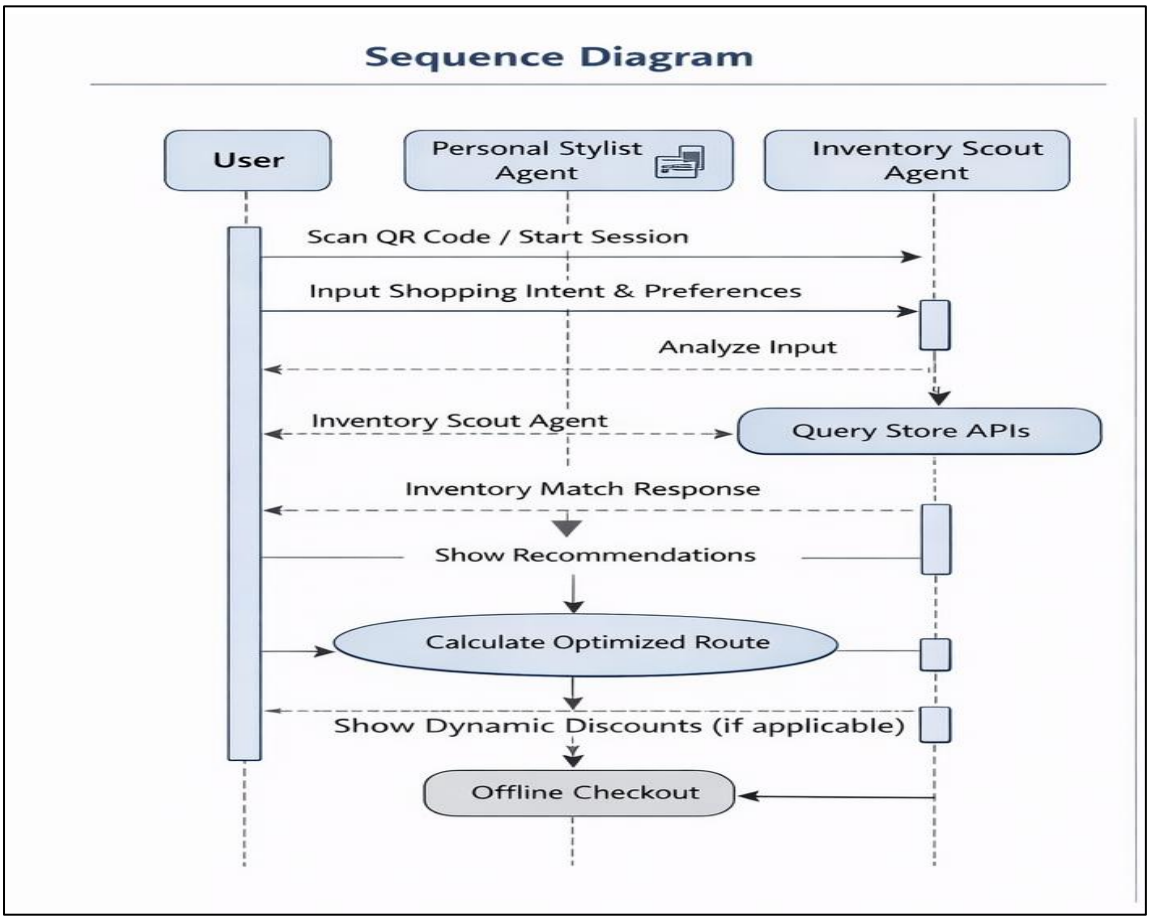
- **Entities**: User, Mall Session, Store, Inventory Item, Discount Coupon, Navigation Path
- **Inventory Schema:** ID, store_id, product_name, category, price, stock_level.
- **User Sessions:** ID, user_id, preferences (from Stylist Agent), current itinerary.

12. Technical Workflow Diagram(s)

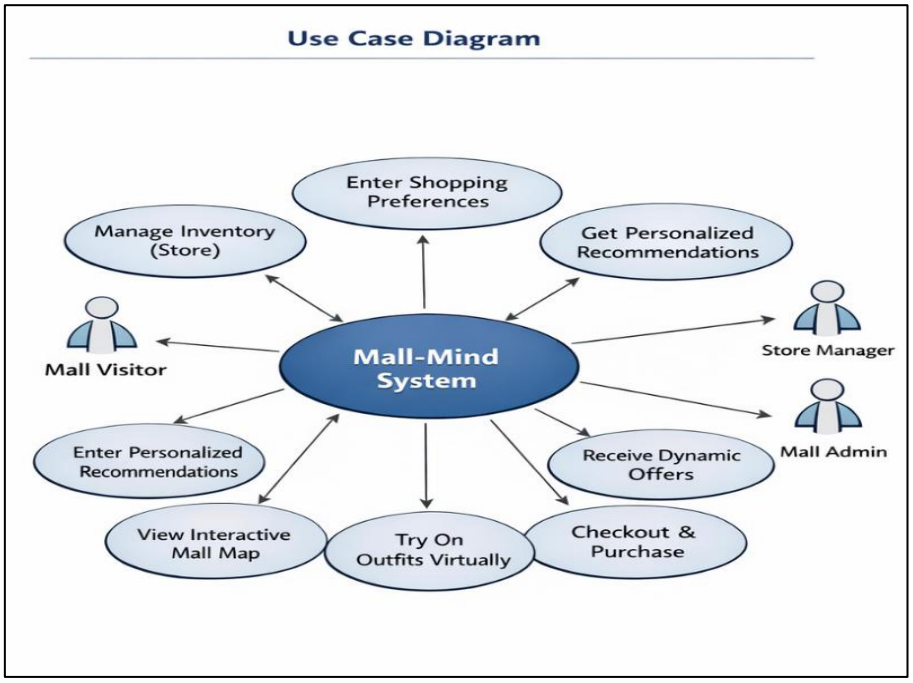
1. State Transition Diagram



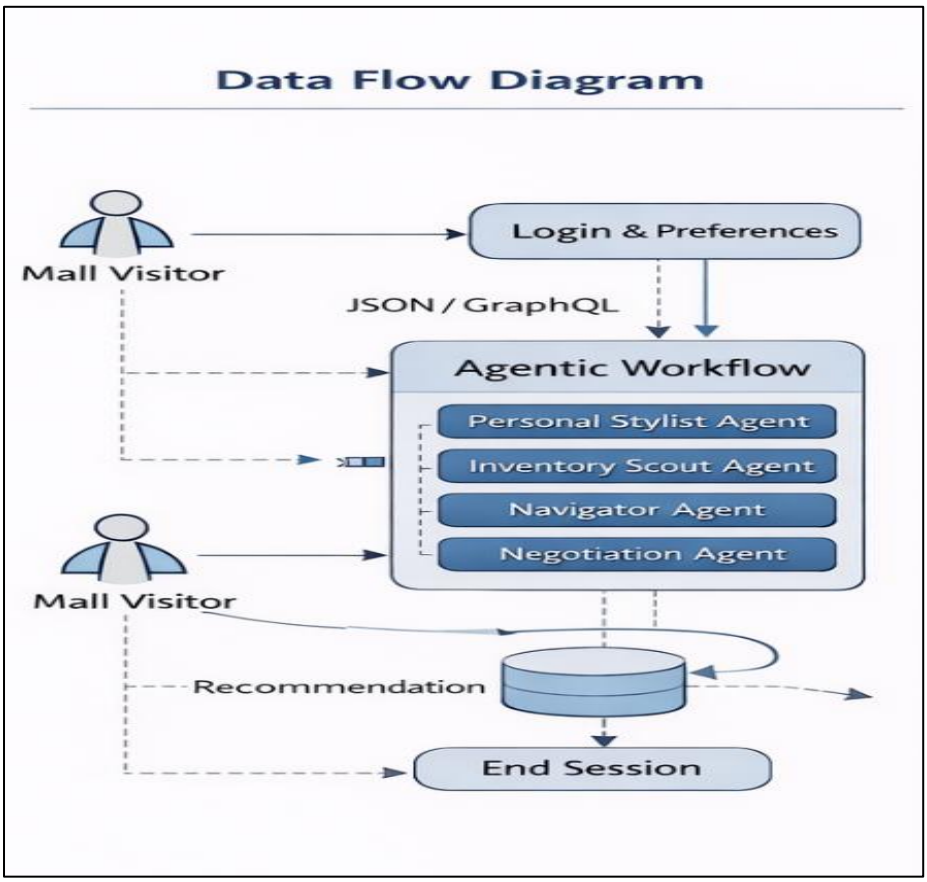
2. Sequence Diagram



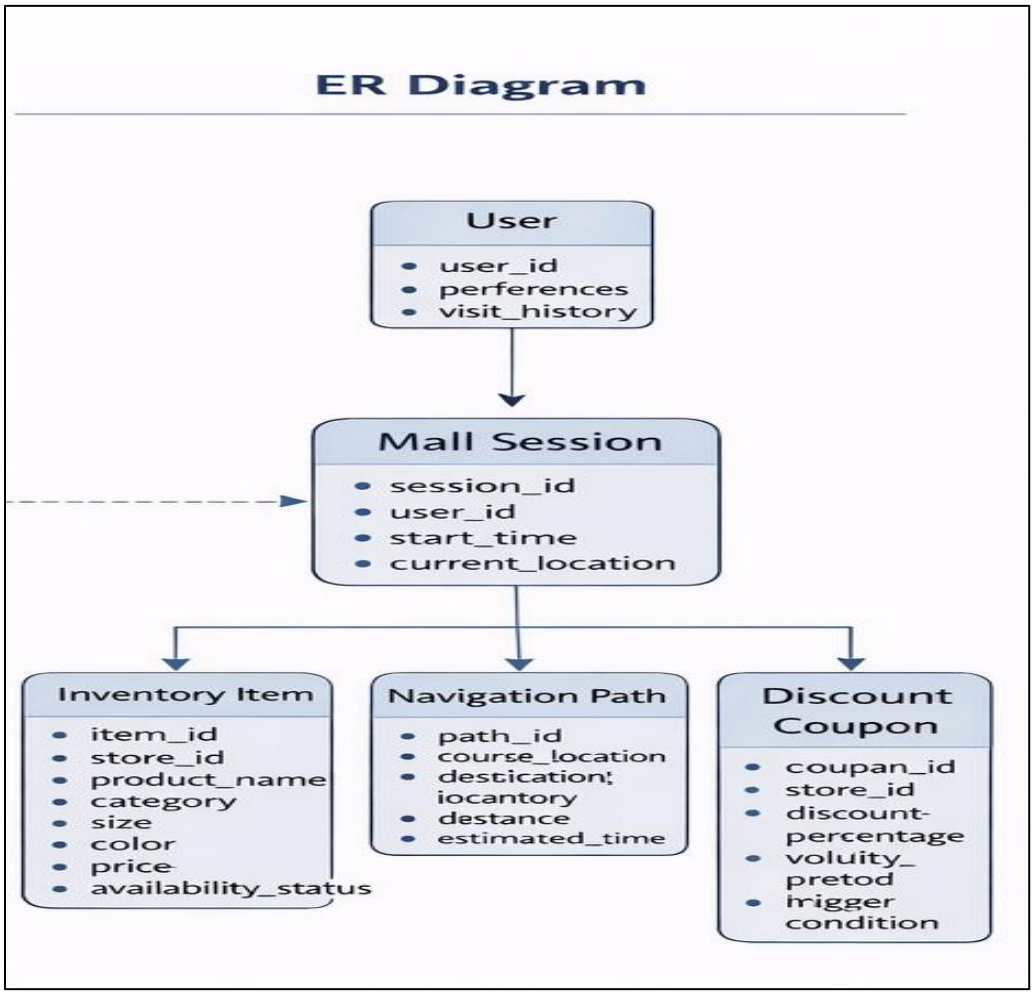
3. Use Case Diagram



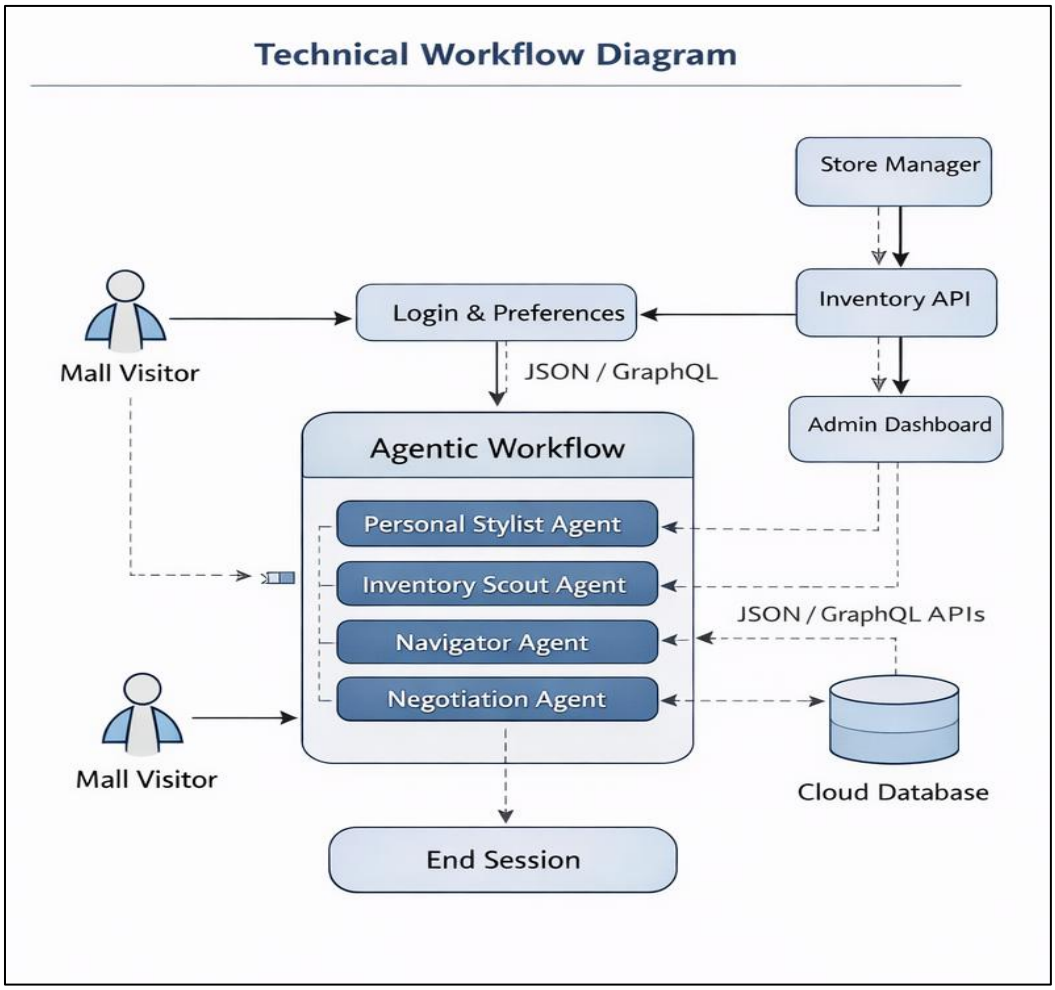
4. Data Flow Diagram



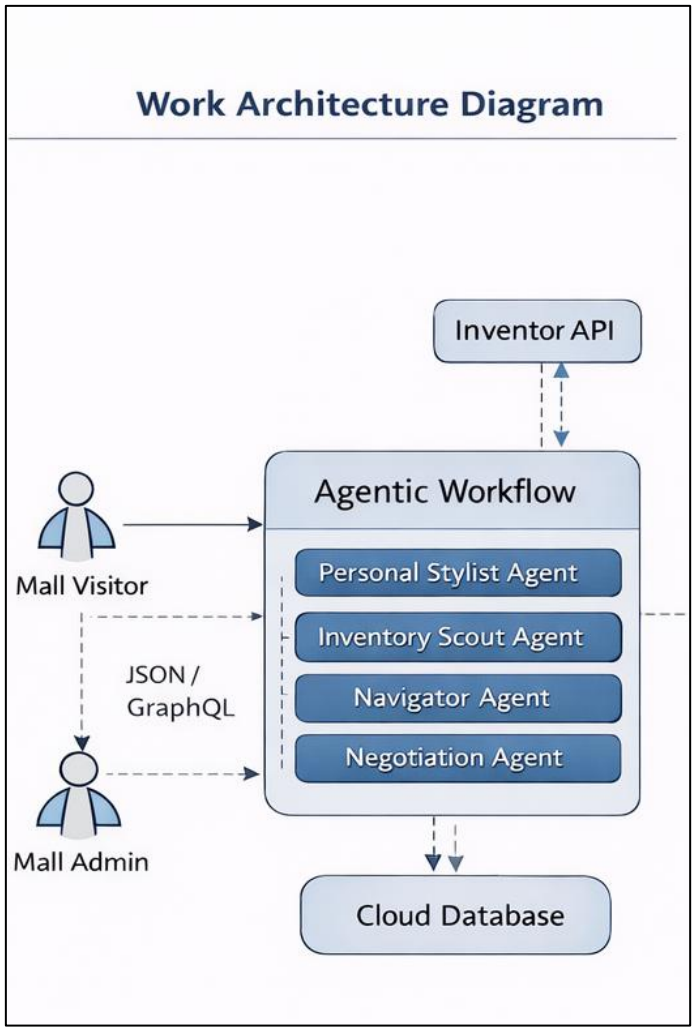
5. ER Diagram


ER Diagram
User
user_id
perferences
visit_history
Mall Session
session_id
user_id
start_time
current_location
Inventory Item
item_id
store_id
product_name
category
size
color
price
availability_status
Navigation Path
path_id
course_location
destication!
iocantory
destance
estimated_time
Discount Coupon
coupan_id
store_id
discount-percentage
voluity_pretod
rigger condition

6. Technical Workflow Diagram



7. Work Architecture Diagram



13. Quality: NFRs and Testing

- Availability: ≥ 99%
- Latency**:** ≤5 seconds
- Error rate: ≤ 1%
- Security**:** JWT authentication and authorisation
- Scalability**:** Multi-store support and concurrent users
- Testing: Unit (backend), E2E (UI), Integration (APIs).

14.Security and Compliance

- JWT-based authentication for secure user sessions
- Temporary and anonymised shopping sessions
- Secure HTTPS communication for all APIs

- Role-based access control for users, stores, and admins
- Uploaded images for virtual try-on were deleted after the session ended

15. Delivery and Operations

- Release Plan: Demo in Week 12.
- CI/CD: GitHub
- Monitoring**:** Logs + performance alerts.

16.Risks & Mitigation

Risk	Probability	Impact	Mitigation
API failure	Medium	High	Mock services
Database issues	Medium	Medium	Optimize queries, use caching
AI inaccuracies	Low	Medium	Continuous retraining
Time constraints	Low	Medium	Weekly reviews

17.Research and Evaluation

- Study of smart retail and mall digitisation systems
- Analysis of e-commerce personalisation techniques
- Research on agent-based AI architectures
- Review of indoor navigation and path optimisation algorithms
- Evaluation based on recommendation accuracy, response time, and user satisfaction

18.Appendices

- **Glossary:** Agentic AI, NLP, JWT, PWA, E2E, AR.
- **References:** Research papers on smart retail systems,OpenAI Assistants API documentation, Indoor navigation and agent-based system studies